

Mathematical Models of Anaerobic Metabolism in Cycling

A literature review focused on separating the phosphocreatine (alactic) and glycolytic (lactic) systems, and modeling their depletion & restoration from power output.

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0. TL;DR — the answer to the core question

Yes, you can model both systems from a power trace, but no single mainstream model does it cleanly. The literature splits into three families, ordered by how explicitly they separate phosphocreatine (PCr) from glycolysis:

Family	Separates PCr vs glycolytic?	Driven by power?	Depletion	Restoration
A. Critical Power / W' -balance	✗ No — lumps all anaerobic work into one tank W'	✓ Yes	linear in $(P - CP)$	mono-exponential, $\tau_{W'}$
B. Hydraulic "tank" models (Margaria–Morton)	✓ Yes — explicit AnF (phosphagen) + AnS (glycolytic) vessels	✓ Yes (power = outflow tap)	fluid drains by vessel	refill via inter-vessel tubes
C. Bioenergetic supply/demand ODEs	✓ Yes — separate alactic / lactic / aerobic metabolic-rate terms	✓ Yes	error-signal ODEs	time-constant + power-gated removal

If your project ("AnaerobicFuelTanks") wants literal, separately-tracked fuel tanks for PCr vs lactic, Family B (the Margaria–Morton hydraulic model, esp. the Weigend/Morton three-component formalization) is the closest published match, and Family C gives you the mechanistic ODEs to make the tanks power-driven and physiologically calibrated.

1. The two anaerobic systems (what we are trying to model)

Classic partitioning of "anaerobic" energy into two sub-systems (Margaria; di Prampero):

Alactic / phosphagen system (ATP–PCr) - Highest metabolic *power* (rate), smallest *capacity* — exhausted in a few seconds of maximal effort. - Physiological store $\approx$ intramuscular PCr ($\sim 20\text{--}25\text{ mmol}\cdot\text{kg}^{-1}$ wet muscle). - Depletion: near-instant at exercise onset / any supra-demand transient. - **Restoration is fast and largely oxidative**: classic alactic O_2 -debt half-time $\approx 30\text{ s}$ (Margaria). ^{31}P -MRS gives PCr-recovery time constants of $\sim 20\text{--}40\text{ s}$, strongly **pH-dependent** (acidosis slows it), and **biphasic** (fast + slow component) under high glycolytic load. - Harris et al. 1976 (the foundational biopsy study) — biphasic PCr resynthesis, fast half-time $\approx 20\text{--}22\text{ s}$, slow component $\sim 170\text{ s}$. - Yoshida et al. 2013, *Scand J Med Sci Sports* — τ_{PCr} differs by muscle (17–43 s), PMID [23662804](#), DOI [10.1111/sms.12081](#). - Intersubject acidosis effect on τ_{PCr} driven by proton-efflux rate — *Am J Physiol Cell Physiol* 2007, DOI [10.1152/ajpcell.00023.2007](#).

Lactic / glycolytic system (anaerobic glycolysis → lactate) - Lower peak power than PCr, much larger capacity; dominant for ~10 s~2 min all-out. - "Fuel" state is best tracked as **accumulated muscle/blood lactate** (a proxy for H⁺ / metabolite accumulation). - Depletion: engaged once demand outstrips aerobic + alactic supply; e.g. glycolysis supplies ~40% of ATP in the *first* of 10×6 s sprints, falling below 10% by the last (Nature Sci Rep 2024, DOI 10.1038/s41598-024-78916-z). - **Restoration is slow**: lactic O₂-debt half-time ≈ ~15 min (Margaria); blood-lactate recovery half-time ~1366 s in the CP literature below.

Reference reviews on this partition - di Prampero & Ferretti, "The energetics of anaerobic muscle metabolism: a reappraisal...", PMID **10647856**, DOI 10.1016/S0034-5687(99)00083-3. - "Measurement of Anaerobic Capacities in Humans" (MAOD as gold standard), *Sports Med* 1993, DOI 10.2165/00007256-199315050-00003. - Alternative single-effort split: alactic (PCr) from post-exercise VO₂ + glycolytic from peak blood lactate — DOI 10.1038/srep42485 (cycling), and conversion-factor method DOI 10.3389/fphys.2023.1147321.

2. Family A — Critical Power & W'-balance (the power-native standard)

What it is. The dominant power-based framework in cycling. Two parameters from a maximal power–duration test: - **CP** (critical power, W) — highest sustainable rate, ~aerobic asymptote. - **W'** ("W-prime", J) — a *single finite* work reserve above CP, i.e. **all anaerobic capacity lumped together** (PCr + glycolytic, not separated).

Core equations. - Power–duration hyperbola: $P = W' / t + CP \rightarrow t_{lim} = W' / (P - CP)$. - Linear work form: $W = W' + CP \cdot t$.

Depletion + reconstitution (this is the depletion/restoration you asked about):

Skiba 2012 — integral "W'bal" model ("Skiba1"):

$$W'_{bal}(t) = W'_0 - \int_0^t W'_{exp}(u) \cdot e^{-(t-u)/\tau_{W'}} du$$

- $W'_{exp} = (P - CP)$ while $P > CP$ (expenditure, linear in power above CP).
- Reconstitution is **mono-exponential** with time constant $\tau_{W'}$, which depends on how far *below* CP you recover ($D_{CP} = CP - P_{recovery}$):

$$\tau_{W'} = 546 \cdot e^{(-0.01 \cdot D_{CP})} + 316 \quad (s)$$

The +316 s asymptote means recovering harder than ~316 W below CP yields no further benefit. Skiba PL et al., *Med Sci Sports Exerc* 2012, PMID **22382171**, DOI 10.1249/MSS.0b013e31824cfdc0.

Skiba 2015 differential form ("Skiba2"): chemical-kinetics derivation letting you compute W'_{bal} continuously / in real time (compartmentalized for $P > CP$ and $P \leq CP$). DOI 10.1249/MSS.0000000000000226.

Froncioni–Clarke–Skiba differential (the GoldenCheetah implementation):

$$\begin{aligned} dW'_{bal}/dt &= -(P - CP) && \text{if } P > CP && \text{(expenditure)} \\ dW'_{bal}/dt &= (CP - P) \cdot (W'_0 - W'_{bal}) / W'_0 && \text{if } P \leq CP && \text{(reconstitution)} \end{aligned}$$

Simple, τ -free, integrates the recovery rate directly from instantaneous power.

Bartram 2018 — elite-athlete recalibration (Skiba2 under-predicts elite reconstitution):

$$\tau_{W'} = 2287.2 \cdot e^{(-0.01 \cdot D_{CP})} + 286.9 \quad (\text{s})$$

DOI 10.1123/ijsp.2017-0356.

Reported reconstitution behaviour (calibration targets): - W' recovered $\approx 37\% / 65\% / 86\%$ after 2 / 6 / 15 min of low-power recovery. - W' reconstitution **half-time** ≈ 234 s — *does not* align with VO_2 half-time (~ 74 s) or blood-lactate half-time (~ 1366 s). This mismatch is itself evidence that a single lumped tank with one τ is physiologically incomplete.

Key limitations (why it motivates a PCr-vs-lactic split): - W' lumps PCr and glycolytic capacity into one reservoir — you cannot read out the two systems separately. - Mono-exponential recovery with a **single τ** cannot capture the observed **slowing of reconstitution across repeated bouts** (a known PCr behaviour), nor the fast/slow biphasic PCr kinetics. - $\tau_{W'}$ derived on **untrained** subjects and is **highly individual**; recommend personalizing. - Caen et al.: faster depletion $\rightarrow$ faster reconstitution despite no VO_2 /lactate/pH differences — mechanism still unresolved.

Best single entry point: Chorley & Lamb narrative review, "Application of CP, W' , and its Reconstitution", [PMC7552657](#), *Sports (Basel)* 2020, DOI 10.3390/sports8090123 — this is where Eqs. 5–9 above are laid out and critiqued.

3. Family B — Hydraulic "tank" models (the explicit PCr-vs-lactic split)

This is the family your repo name evokes. Energy systems are literal *vessels of fluid*; power output is a **tap** draining fluid; recovery is fluid **flowing back between vessels through connecting tubes**. Crucially it has **two separate anaerobic tanks**.

Lineage - Margaria (1976) — original hydraulic analogy of the three energy systems. - **Morton, R.H. (1986)** "A three component model of human bioenergetics", *J Math Biol*, DOI 10.1007/BF01236892 — formalized three interconnected vessels; also "Modelling human power and endurance", DOI 10.1007/BF00171518. - **Sundström** — applied/validated hydraulic-analogy models (sprint roller skiing), showing they generalize across sports. - **Weigend, Behncke & Skiba (2021)** — "A new pathway to approximate energy expenditure and recovery of an athlete" (arXiv 2104.07903) and the **three_comp_hyd** open-source implementation ([github.com/faweigend/three_comp_hyd](#)). Removes ties to concrete metabolic measures and **fits the 8 parameters per-athlete by evolutionary computation**. - **Weigend et al. 2021** — "A hydraulic model outperforms work-balance models for predicting recovery kinetics from intermittent exercise" (arXiv 2108.04510) — direct head-to-head vs W' bal.

The three vessels 1. **Ae** — aerobic / oxidative (effectively large-capacity source, feeds the anaerobic tanks). 2. **AnF** — *anaerobic fast* = **phosphagen / PCr** tank (small, drains & refills fast). 3. **AnS** — *anaerobic slow* = **glycolytic / lactic** tank (larger, slower).

Mechanics (conceptual equations). Each vessel has a capacity (cross-section $\times$ height) and liquid level h . Power demand $p(t)$ sets the outflow. Flow between vessels is proportional to the **difference in liquid levels** and gated by **tube height/position parameters** — so a tank only refills once the level it draws from is high enough. The Weigend abstraction uses **~8 parameters**: { AnF capacity, AnS capacity, Ae max flow (M_{Ae}),

tube position ϕ (height of the Ae→AnF/AnS outlet), γ (AnS tube height), θ (positions/offsets of the anaerobic tank tops) }, fitted per athlete.

Why this family answers your question best - Depletion is separated by construction: PCr (AnF) drains first and fastest on any power transient; glycolytic (AnS) engages as AnF empties — reproducing the physiological sequencing without you hard-coding it. - **Restoration is naturally multi-rate:** AnF refills quickly from Ae/AnS via its tube; AnS refills slowly — giving the fast/slow, power-dependent recovery that Family A's single τ misses. Weigend 2021 shows this **out-predicts W'bal for intermittent recovery kinetics**. - **Fully power-driven:** the only time-series input is the power demand trace.

Trade-off: parameters are abstract (fitted, not directly measured), so mapping tank levels back to "mmol PCr" or "mmol lactate" requires an extra calibration step (see §5).

4. Family C — Bioenergetic supply/demand ODE models (mechanistic, from power)

These write **metabolic rate** as *demand = supply*, with supply split into **explicit alactic, lactic, and aerobic terms** — the most physiologically direct "PCr vs lactic from power" formulation.

Lineage: di Prampero's energetics → Artiga Gonzalez et al. (VO₂ from power) → the intermittent-cycling model below.

Anchor paper (has a full state-space ODE system): "Development and validation of a dynamic bioenergetic model for intermittent ergometer cycling", *Eur J Appl Physiol* 2023, PMID 37369795, DOI 10.1007/s00421-023-05256-7 (PMC10638188). 14 trained cyclists; 17 parameters (14 fit by grey-box `nlgreyest`). Structure:

- **Demand** = principal cycling work $MR_{work} = a + b \cdot P$ (linear in measured power) + ventilation cost (~11% of MR) + accumulated-metabolite cost (~6.9% of MR) + resting metabolism.
- **Supply** = alactic + lactic + aerobic:
- **Aerobic** — mono-exponential toward demand: $d(MR_{aer})/dt = (MR_{demand} - MR_{aer}) / \tau_{aer}$, with $\tau_{aer} \approx 25$ s (constrained 10–100 s, per di Prampero).
- **Lactic (glycolytic)** — same error-signal/time-constant form but the error subtracts *both* current aerobic and lactic rates, and is damped by $0.8 \times d(MR_{aer})/dt$; $\tau_{la} \approx 12.5$ s (constrained 10–15 s, from Gastin's kinetics). Constrained ≥ 0 .
 - **Fuel state = normalized muscle lactate** $[mLa] \in [0, 1]$ (0 = fully recovered, 1 = max accumulation).
 - **Depletion:** $[mLa]$ accumulates at the (normalized) lactic rate.
 - **Restoration is power-gated:** removal $\propto [mLa] \times demand_factor$, where $demand_factor = 1$ at rest and **falls linearly to 0 at the lactate threshold (LT1)**. → lactate clears fastest at low power and **stops clearing above LT1**. This is exactly a *power-driven* restoration law for the glycolytic tank.
- **Alactic (phosphocreatine)** — an **instantaneous buffer**: it supplies whatever demand aerobic + lactic cannot, capped by $MR_{max} - MR_{aer}$.
 - **Fuel state = normalized alactic depletion** $\in [0, 1]$ (0 = full PCr, 1 = empty).
 - **Restoration with a hysteresis/efficiency cost:** when the alactic rate goes *negative* (during low-power phases) the tank refills, but only a fraction η of the diverted energy becomes usable alactic store — cost

$(1-\eta)$. This is the model's PCr-recovery law.

Reported outputs (calibration targets): anaerobic contribution ≈ 59.6 kJ / 75.6 kJ across protocols; in the harder protocol the alactic state overshoot to 1.12 and [mLa] to 1.21 (i.e. the P2-fit capacities were exceeded by 12% / 21%).

Stated limitations: no aerobic **slow component** ($\rightarrow$ under-predicts aerobic MR, over-predicts anaerobic in long low-intensity recovery); a two-compartment lactate model and an [mLa]-dependent τ were tried but made the system underdetermined — a caution for your own parameterization.

Related dynamic model: the **bi-exponential** reconstitution/expenditure of W' in trained cyclists, *Eur J Sport Sci* 2023, DOI 10.1080/17461391.2023.2238679 — a bridge between Family A and the fast/slow (PCr-like vs glycolytic-like) two-component idea.

5. Synthesis — how to actually model PCr vs lactic from power

A pragmatic recipe combining the families, matched to "depletion & restoration":

1. **Two anaerobic tanks, power as the only time-series input** (Family B geometry, Family C laws): - **PCr tank (AnF)**: small capacity ($\sim$ PCr store); drains first on any demand > aerobic supply; refills fast, oxidatively, **efficiency-limited** (η hysteresis cost), with a **biphasic / pH-slowed** rate ($\tau_{\text{fast}} \sim 20\text{--}30$ s, slowed by acidosis / repeated bouts). - **Glycolytic tank (AnS)**: larger capacity; engages as PCr empties; **fuel state tracked as accumulated lactate [mLa]**; refills **slowly and power-gated** — fastest at rest, ceasing above LT1 (Family C law); half-time on the order of minutes.
2. **Aerobic supply as a mono-exponential toward demand** ($\tau_{\text{aer}} \approx 25$ s) sets how much of the demand the anaerobic tanks must cover — the coupling that makes depletion power-dependent.
3. **Depletion law:** at each instant, demand (P) is met by aerobic first, then PCr (instant buffer), then glycolytic — draining tanks in that order.
4. **Restoration laws (the part single- τ models get wrong):** - PCr: fast, oxidative, pH/efficiency-limited, and **slows across repeated bouts**. - Glycolytic: slow, and **only below LT1** (power-gated removal).
5. **Validate against** the CP/ W' reconstitution targets (37/65/86% at 2/6/15 min; half-time ~ 234 s) and, if you have lab data, 31P-MRS τ_{PCr} and blood-lactate kinetics.

Reference implementations to start from: `three_comp_hyd` (Weigend, Python, evolutionary fitting) for the tank machinery; the *EJAP 2023* bioenergetic ODEs (§4) for the power $\rightarrow$ system laws; GoldenCheetah's Froncioni–Clarke–Skiba integrator (§2) for a simple W' baseline to benchmark against.

6. Annotated key references

Critical Power / W' (Family A) - Skiba et al. 2012, *MSSE* — original W'_{bal} integral + $\tau_{\text{W'}}$ = $546 \cdot e^{-(0.01 \cdot D_{\text{CP}})} + 316$. **PMID 22382171**, DOI 10.1249/MSS.0b013e31824cfdc0. - Skiba et al. 2015, *MSSE* — differential W'_{bal} . DOI 10.1249/MSS.0000000000000226. - Bartram et al. 2018, *IJSP* — elite $\tau_{\text{W'}}$ = $2287.2 \cdot e^{-(0.01 \cdot D_{\text{CP}})} + 286.9$. DOI 10.1123/ijsp.2017-0356. - Chorley & Lamb 2020, *Sports* — **review of all of the above**, PMC7552657, DOI 10.3390/sports8090123. - Clarke & Skiba 2013, *Adv Physiol Educ* — rationale / teaching of the power–duration model.

Hydraulic tank models (Family B) - Morton 1986, *J Math Biol* — three-component model. DOI 10.1007/BF01236892. - Morton, "Modelling human power and endurance", *J Math Biol*. DOI 10.1007/BF00171518. - Weigend, Behncke, Skiba 2021 — abstract three-component hydraulic model, arXiv 2104.07903; code: github.com/faweigend/three_comp_hyd. - Weigend et al. 2021 — hydraulic vs W'bal for recovery kinetics, arXiv 2108.04510. - Sundström et al. 2021 — hydraulic-analogy validity/reliability (roller skiing), ResearchGate 354553483. - "Individualized physiology-based digital twin... reinterpretation of Margaria–Morton", *Sci Rep* 2024, DOI 10.1038/s41598-024-56042-0 (PMC10915161).

Bioenergetic supply/demand ODEs (Family C) - Dynamic bioenergetic model, intermittent cycling, *EJAP* 2023 — **PMID 37369795**, DOI 10.1007/s00421-023-05256-7 (PMC10638188). (*Primary mechanistic source for §4.*) - Bi-exponential reconstitution/expenditure of W', *EJSS* 2023, DOI 10.1080/17461391.2023.2238679. - Evaluating bioenergetic pathway contributions single → multiple sprints, *Sci Rep* 2024, DOI 10.1038/s41598-024-78916-z. - Dynamic model of bioenergetics / grey-box VO₂-from-power (Artiga Gonzalez et al.).

PCr / glycolytic depletion & restoration kinetics (physiology to calibrate against) - Harris et al. 1976 — biphasic PCr resynthesis (fast $t_{1/2} \approx 20\text{--}22$ s), *Pflügers Arch.* - Yoshida et al. 2013, *Scand J Med Sci Sports* — τ_{PCr} by muscle, **PMID 23662804**, DOI 10.1111/sms.12081. - Proton-efflux / acidosis effect on τ_{PCr} , *AJP Cell* 2007, DOI 10.1152/ajpcell.00023.2007. - di Prampero & Ferretti 1999 — anaerobic energetics reappraisal, **PMID 10647856**, DOI 10.1016/S0034-5687(99)00083-3. - MAOD / anaerobic capacity measurement, *Sports Med* 1993, DOI 10.2165/00007256-199315050-00003; single-effort cycling split DOI 10.1038/srep42485.

Notes on sourcing: full text of PMC10638188 and PMC7552657 was retrieved via PubMed Central; their typeset equations were embedded as images, so the closed forms in §2 and §4 are the standard published forms cross-checked against the surviving prose (variable definitions, the "+316 W" asymptote, the τ constraints and values). Verify exact coefficients against the publisher PDFs before hard-coding.